Time-Series Forecasting with ARIMA

Trend • Seasonality • Stationarity • 30-Day Forecast

Project report | Supplied daily-demand-series.csv | 1–30 June 2026

1. Executive Summary

The supplied dataset has 30 daily observations from 2026-06-01 through 2026-06-30. This project decomposes demand, tests stationarity using ADF, selects ARIMA by chronological holdout RMSE, evaluates with RMSE and MAPE, and forecasts 1–30 July 2026. Selected model: ARIMA(0, 2, 2). Holdout RMSE=12.91; MAPE=6.55%; MAE=10.51.

2. Dataset and Preparation

The CSV contains date, demand, marketing_event and holiday. Date was parsed, sorted and indexed daily. The first 24 days are training and the final six days are a chronological holdout. Marketing-event and holiday columns are not used as regressors because future values are unavailable.

3. Decomposition

Additive decomposition with period=7 separates observed demand into trend, weekly seasonal and residual components. The weekly period is a reasonable hypothesis, but only about four cycles are present, so seasonal estimates are exploratory.

4. Stationarity and Differencing

ADF on training levels: statistic=-1.806, p=0.378. ADF on first differences: statistic=-1.291, p=0.633. Both p-values exceed 0.05, so the unit-root null is not rejected. With only 24 observations, the test has low power. Candidate d values 0, 1 and 2 were compared; the selected ARIMA order has d=2.

5. ARIMA Model Selection

A grid over p=0..2, d=0..2 and q=0..2 was fitted on the first 24 observations. Each candidate forecast the six-day holdout; the lowest RMSE selected ARIMA(0, 2, 2). Holdout RMSE=12.913; MAPE=6.549%; MAE=10.514.

6. Thirty-Day Forecast

The selected order was refitted on all 30 days and forecasts the next 30 days, 1–30 July 2026. Point forecast begins at 149.1 and ends at 185.9. Prediction intervals widen with horizon and must be considered.

7. Limitations

Only one month of data is available; weekly seasonal estimates and a 30-day projection are therefore uncertain. ADF has low power on a short sample. Future holiday and marketing-event flags are unknown. With more history, use rolling-origin validation, residual diagnostics, and compare SARIMA/Prophet and seasonal-naive baselines.

8. Conclusion

The complete requested pipeline is implemented. ARIMA(0, 2, 2) achieved holdout RMSE 12.91 and MAPE 6.55%. Treat this as a reproducible baseline rather than a high-confidence operational forecast until more historical data are collected.

9. Reproducibility

The accompanying notebook contains data loading, decomposition, ADF tests, ARIMA grid search, chronological holdout evaluation, final forecast, charts, and CSV export. Keep daily-demand-series.csv in the same directory. Install pandas, numpy, matplotlib, statsmodels, scikit-learn and jupyter.

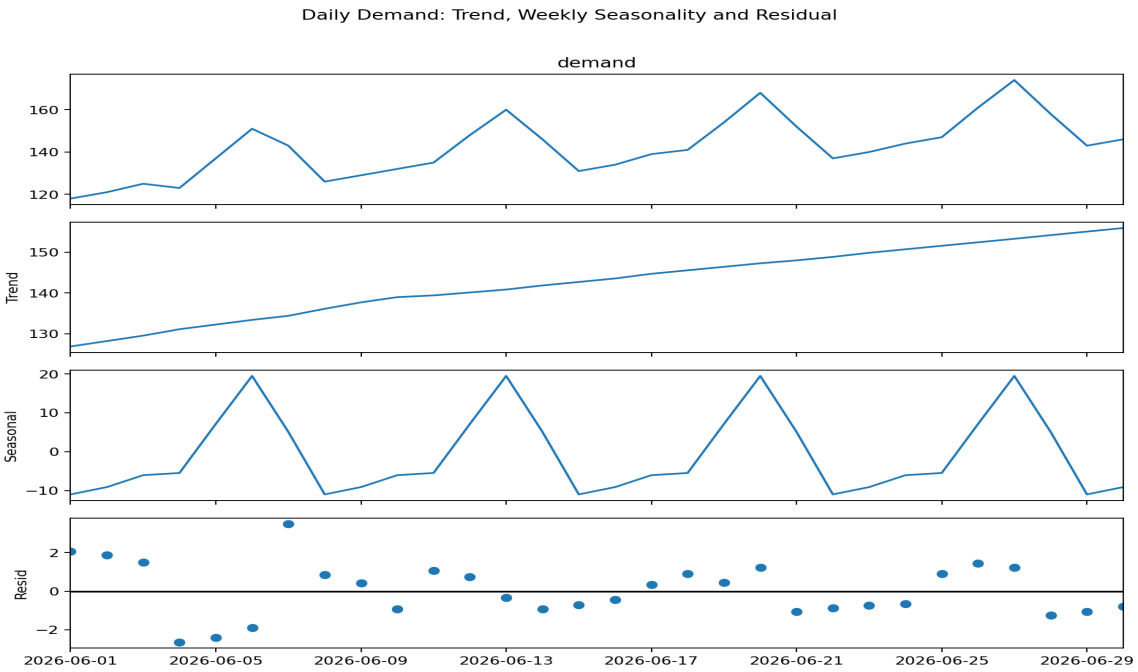


Figure 1. Additive trend, weekly seasonality and residual.

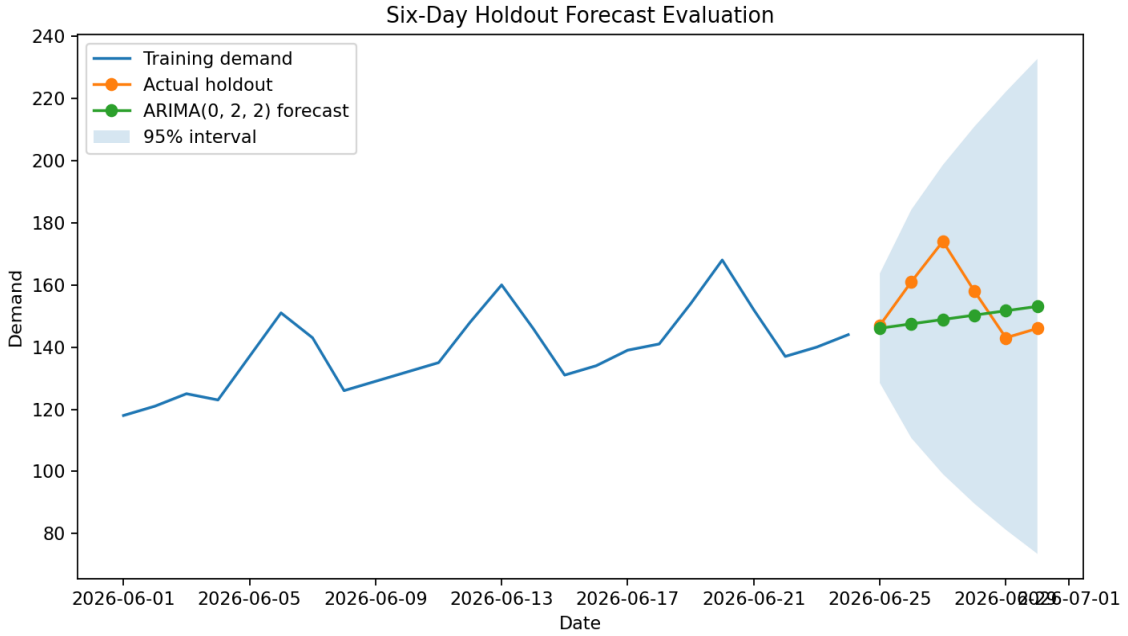


Figure 2. Six-day holdout prediction.

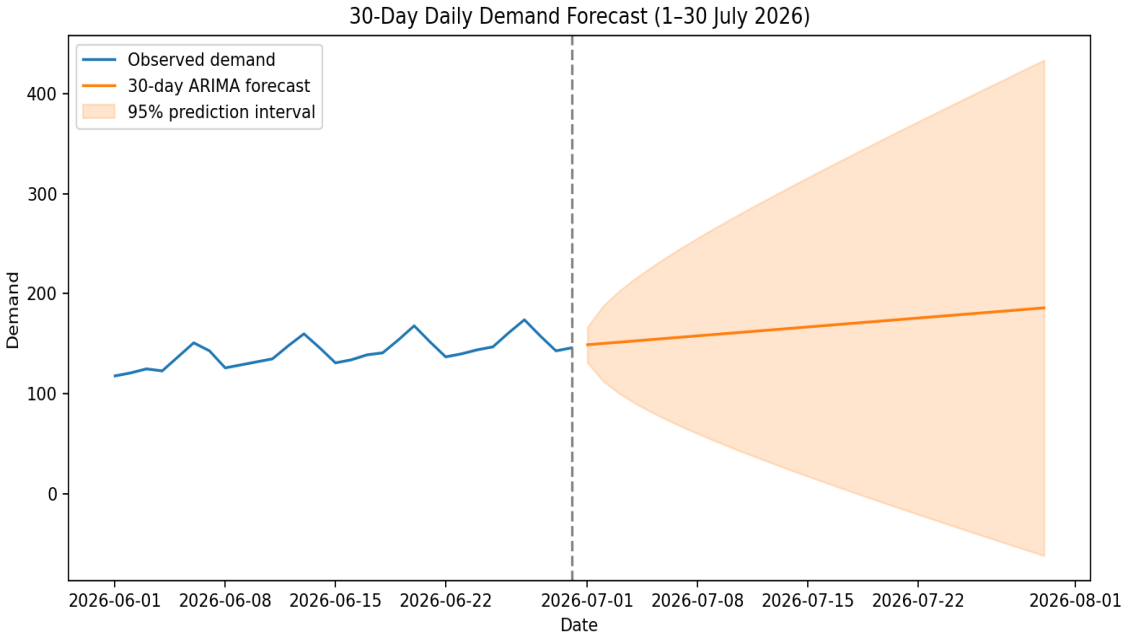


Figure 3. Thirty-day forecast and 95% prediction interval.